

THE UNIVERSITY OF EDINBURGH

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**Optimization of Bike Rebalancing: Quantifying the Trade-offs Between
Fleet Size and Rebalancing Vehicle Capacity for Emission Reduction**

Methodology, Modelling and Consulting Skills
Goup Number 03

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1 Introduction

The project touches upon the optimal daily operations of the bike station rebalancing crew. The rebalancing crew consists of a certain number of vehicles with a limited capacity. The rebalancing occurs twice a day — before the operational day begins and during the day (e.g., at 2 pm). The decision support is based on estimated demand data in 30-minute slots, which is grouped into a certain number of scenarios, for instance, based on the weather conditions. The first rebalancing of the day has to be decided beforehand, without knowing which scenario will come true. However, the second rebalancing is done after the scenario realization, and therefore, the demand is treated as certain. The goal of our model is to achieve a sustainable operations plan, minimizing the total emissions.

It is assumed that unfulfilled demand is fulfilled by a car, which incurs appropriate CO2 emissions with values determined by the trip distance, and the average emissions per km by a car. The demand can be unfulfilled when either there is no bike at a starting station or there is no place left at the ending station, which encourages appropriate rebalancing to keep the stations operational. Additionally, we require that certain stations (e.g., those in commercial areas) have a minimum percentage of demand that has to be served throughout the day. Such a constraint must be fulfilled with a certain probability.

However, the rebalancing requires us to drive a van, which also incurs emissions. A simplified model of vehicle routing is assumed, where the total distance traversed by our cars is the sum of the distances between all bike transports between the stations. Thus, the second element in the objective is the cost minimization, where the costs are measured as the total distance traveled by vehicles multiplied by the average van emissions per km.

To quantify the robustness of our model, an out-of-scenario validation technique is performed. This allows us to gain better insight into whether our model accurately prepares for the unexpected changes in demand, which might cause the demand at the commercial station to be unfulfilled with more probability than previously expected.

2 Modeling

2.1 Notation

The notation used can be found in Table 1.

Sets		
Ω		Stochastic scenarios
L		Stations
L^*		Commercial stations
P^D		Pairs of stations i, j , such that there is a nonzero demand between them
$P^D(i, :)$	$i \in L$	Stations j , such that $(i, j) \in P^D$
$P^D(:, i)$	$i \in L$	Stations j , such that $(j, i) \in P^D$
T		Time points
T^R		Time points when rebalancing takes place
T^N		Time points when rebalancing does not take place
T_0		Time points with the initial point
Variables		
x_{ijtw}	$(i, j) \in P^D \ t \in T \ \omega \in \Omega$	Number of riders travelling from station i to station j at time t in scenario ω
I_{itw}	$i \in L \ t \in T_0 \ \omega \in \Omega$	Number of bikes in docks of station i at the end of time t in scenario ω
y_i	$i \in L$	Initial allocation of the bikes at station i , before the operational phase starts
r_{ijtw}	$i, j \neq i \in L \ \forall t \in T^R \ \omega \in \Omega$	Number of bikes rebalanced from station i to station j
z_ω	$\omega \in \Omega$	Indicator if the minimum demand constraint is fulfilled in scenario ω
Parameters		
F		Total fleet of bikes available
g_i	$i \in L$	Capacity of station i
d_{ijtw}	$(i, j) \in P^D \ t \in T \ \omega \in \Omega$	Demand for going from station i to station j at time t
v		Rebalancing capacity of all vehicles available
h_{ij}	$i \in L, j \in L$	Time it takes to travel from station i to station j on bike
s_{ij}	$i \in L, j \in L$	Driving distance between stations i and j
e^C		CO2 emissions per km travelled by car
e^V		CO2 emissions costs per km when using a transportation van
p		Minimum outgoing commercial station demand that has to be served
q		Required probability of the minimum level of demand being fulfilled

Table 1: Notation used

2.2 Model

$$\min \frac{1}{|\Omega|} \cdot \sum_{\omega \in \Omega} \left[e^C \cdot \sum_{(i,j) \in P^D} \sum_{t \in T} s_{ij} \cdot (d_{ijtw} - x_{ijtw}) + e^V \cdot \sum_{i,j \neq i \in L} \sum_{t \in T^R} s_{ij} \cdot r_{ijtw} \right] \quad (1)$$

s.t.

$$\sum_{i \in L} y_i \leq F \quad (2)$$

$$y_i \leq g_i \quad \forall i \in L \quad (3)$$

$$\sum_{\omega \in \Omega} z_\omega \geq q \cdot |\Omega| \quad (4)$$

$$I_{i,0,\omega} = y_i \quad \forall i \in L \quad \forall \omega \in \Omega \quad (5)$$

$$\sum_{i \in L^*} \sum_{t \in T} \sum_{j \in P^D(i,:)} x_{ijtw} \geq z_\omega \cdot p \cdot \sum_{i \in L^*} \sum_{t \in T} \sum_{j \in P^D(i,:)} d_{ijtw} \quad \forall \omega \in \Omega \quad (6)$$

$$I_{itw} \leq g_i \quad \forall i \in L \quad \forall t \in T \quad \forall \omega \in \Omega \quad (7)$$

$$\sum_{j \neq i \in L} r_{ijtw} + \sum_{j \in P^D(i,:)} x_{ijtw} \leq I_{i,t-1,\omega} \quad \forall i \in L \quad \forall t \in T^R \quad \forall \omega \in \Omega \quad (8)$$

$$\sum_{j \in P^D(i,:)} x_{ijtw} \leq I_{i,t-1,\omega} \quad \forall i \in L \quad \forall t \in T^N \quad \forall \omega \in \Omega \quad (9)$$

$$x_{ijtw} \leq d_{ijtw} \quad \forall (i,j) \in P^D \quad \forall t \in T \quad \forall \omega \in \Omega \quad (10)$$

$$\sum_{i,j \neq i \in L} r_{ijtw} \leq v \quad \forall t \in T^R \quad \forall \omega \in \Omega \quad (11)$$

$$I_{itw} - I_{i,t-1,\omega} = \sum_{j \in P^D(:,i) | t > h(j,i)} x_{ji,t-h(j,i),\omega} - \sum_{j \in P^D(i,:)} x_{ijtw} \quad \forall i \in L \quad \forall t \in T^N \quad \forall \omega \in \Omega \quad (12)$$

$$I_{itw} - I_{i,t-1,\omega} = \sum_{j \neq i \in L} (r_{jitw} - r_{ijtw}) + \sum_{j \in P^D(:,i) | t > h(j,i)} x_{ji,t-h(j,i),\omega} - \sum_{j \in P^D(i,:)} x_{ijtw} \quad \forall i \in L \quad \forall t \in T^R \quad \forall \omega \in \Omega \quad (13)$$

with domain constraints:

$$y_i \in \mathbb{Z}_+ \quad \forall i \in L \quad (14)$$

$$z_\omega \in \{0, 1\} \quad \forall \omega \in \Omega \quad (15)$$

$$x_{ijtw} \in \mathbb{Z}_+ \quad \forall (i,j) \in P^D \quad \forall t \in T \quad \forall \omega \in \Omega \quad (16)$$

$$r_{ijtw} \in \mathbb{Z}_+ \quad \forall i, j \neq i \in L \quad \forall t \in T^R \quad \forall \omega \in \Omega \quad (17)$$

$$I_{itw} \in \mathbb{Z}_+ \quad \forall i \in L \quad \forall t \in T_0 \quad \forall \omega \in \Omega \quad (18)$$

- Objective (1) minimises the expected emissions over all scenarios. CO2 emissions comprise car emissions, resulting from unfulfilled demand, and van emissions, from the rebalancing phase
- Constraint (2) assures that we won't allocate more bikes than the available fleet
- Constraint (3) makes sure that the initial allocation does not exceed the station capacity
- Constraint (4) acts as a chance constraint, setting the minimum probability of fulfilling the commercial station demands
- Constraint (5) links the first stage variables and second stage ones, by making sure the initial allocation is equal to the initial state of the docks
- Constraint (6) determines if the minimum percentage of demand outgoing from the commercial stations is satisfied
- Constraint (7) controls the dock state to not exceed the station capacity
- Constraints (8) and (9) guarantee that no more bikes can be taken from a station that are already stationed there. This is to prevent a situation where the rider/rebalancing crew would have to get the bike directly from the new incoming rider
- Constraint (10) limits the ridership to the demand

- Constraint (11) guarantees that we won't exceed the rebalancing crew resources
- Constraints (12) and (13) define the change in the dock state, as the number of bikes incoming/rebalanced into from which the number of bikes outgoing/rebalanced from are subtracted

Our rebalancing model is simplified in the sense that no routing is performed, and the rebalancing is always finished before the next time slot. Moreover, the emissions penalty is applied per every bicycle transferred, not per every route taken; this is because, without routing, there is no split of bikes between the vans, and thus a single-trip penalty could be an underestimation, as a single van would need to carry more bikes than its capacity.

Note that Domain Constraint (18) can be relaxed, as the initial dock state will always be an integer, and all of the changes in the dock states will be integers.

2.3 Model validation

As our model operates in a stochastic environment, where demands are uncertain, we must take appropriate measures to ensure the robustness and representativeness of the returned solutions.

We assume that the scenario generation is in the form of a black box, and thus we do not have access to, e.g., CDF functions. However, we can produce any desired number of samples.

Thus, we have to decide on the proper number of scenarios used. On one hand, we can increase the number of scenarios, making the decision space more representative of the real situation. On the other hand, our model grows in size proportionally to the number of scenarios used. This, combined with the fact that our model is a discrete ILP, and thus NP-Hard, we can sustain a major drop in the performance of the model when dimensioning the scenarios.

Our solution is to iteratively grow the number of scenarios in our model while performing an out-of-sample validation procedure. This entails freezing the first-stage variable values and solving the model for a larger number of unseen scenarios, which can be done independently of other scenarios. This can help us measure how many samples we need to make our decisions feasible on the whole distribution (with respect to minimum demand levels fulfilled). Moreover, we can find out whether the approximate expected costs correspond to the real expected costs associated with our solutions. The algorithm pseudocode can be found in Listing 1.

Additionally, we might want to consider the value of perfect information. This is done by designating one of the distribution samples as the "real" one, and solving the model for only this scenario with frozen and free values of the first-stage variables. This gives us information on how sustainable our model can be in practice, and how much of an improvement we can obtain if we have better demand prediction methods.

Algorithm 1 Validation

Require: Number of scenarios S , number of validation scenarios V

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1:  $\Omega \leftarrow \{\text{create\_demand\_scenario}(), \quad k = 1 \dots S\}$ 
2: Solve the problem (1)-(18)
3: Save the 1st stage variable values  $y_i^* \leftarrow y_i, \quad i \in L$ 
4: feasibility_probability  $\leftarrow 0$ 
5: avg_emissions  $\leftarrow 0$ 
6: for  $k \in 1 \dots V$  do
7:    $\Omega \leftarrow \{\text{create\_demand\_scenario}()\}$ 
8:   Solve the the problem (1)-(18) with fixed variable values  $y_i = y_i^*, \quad i \in L$ , and  $z = 1$ 
9:   if The problem was feasible then
10:     feasibility_probability  $+= \frac{1}{V}$ 
11:     avg_emissions  $+= \frac{1}{V} \cdot \text{emissions}$ 
12:   else
13:     Solve the the problem (1)-(18) with fixed variable values  $y_i = y_i^*, \quad i \in L$ , and  $z = 0$ 
14:     avg_emissions  $+= \frac{1}{V} \cdot \text{emissions}$ 
15:   end if
16: end for
17: return feasibility_probability, avg_emissions

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3 Edinburgh case study

We assumed that cars on average emit $0.1294 \frac{\text{kg CO}_2}{\text{km}}$, and vans emit $0.1854 \frac{\text{kg CO}_2}{\text{km}}$. The demand fulfillment goals were set to 50%, which has to be satisfied with the probability of 75%. Moreover, we assumed that the rebalancing takes place exactly once – in the middle of the operational period.

Trips and Stations Data Preparation Microsoft Excel and Power Query were used to consolidate the shared cycle hire trip details. Distances for reported trips were estimated using the average duration and an assumed 25 km/h speed, consistent with UK electric bike regulations. In contrast, unreported trip (lacking historic trip data between stations) distances were calculated using the Law of Cosines formula:

$$d = R \cdot \arccos(\sin \phi_1 \cdot \sin \phi_2 + \cos \phi_1 \cdot \cos \phi_2 \cdot \cos(\lambda_2 - \lambda_1))$$

from which duration was derived using the same speed. Station categorisation was determined by matching stations to the closest point of interest (hospital, residential, university, library, commercial, school) within an assumed 400 m walkable radius, also calculated using the Law of Cosines distance formula. Any missing station capacity was assigned an estimated value equal to the average capacity of the shared station data, which was 25 bikes.

Demand data simulation: The demand data was simulated using a Poisson distribution, with the parameters probed using the demand data from previous years. If we denote the base (past) demand data by $\bar{d}$, we obtain the new demands based on the formula

$$\lambda_t \leftarrow \frac{1}{|P^D|} \sum_{(i,j) \in P^D} \bar{d}_{ijt} \quad \forall t \in T \quad (19)$$

$$d_{ijt} \sim \bar{d}_{ijt} + \mathcal{P}(1.3\lambda_t) \quad \forall (i,j) \in P^D : i \notin L^*, \forall t \in T \quad (20)$$

$$d_{ijt} \sim \bar{d}_{ijt} + \mathcal{P}(3.8\lambda_t) \quad \forall (i,j) \in P^D : i \in L^*, \forall t \in T \quad (21)$$

Where the exact multipliers were fitted to make sure that the stations can be made full, and not all demand can be satisfied (the original data suggested a ridership level, where the commercial demand could always be satisfied).

Sensitivity analysis: We decided to measure the impact of the available fleet of bikes and the rebalancing capacity on the model stability and total emissions. The tests were carried out on all combinations of fleet counts $F \in \{160, 180, 200\}$, and available rebalancing of either 0 (no rebalancing capacity), 12 (the capacity of a single van), and equal to the fleet count (enough cars to rebalance the whole system). The values of fleet counts were selected, as rounded, low values that enable the model feasibility with rebalancing ($F = 160$, and without rebalancing $F = 180$).

Value of perfect information: The value of perfect information was calculated by assigning one of the model's scenarios as the "real" one, and then measuring the difference between the model total emissions when only this scenario was presented to the model vs the case when model has to rely on the maximum number of scenarios that were available to it.

Computational details We run tests using the base demand from past data from June. To measure the robustness of the model, up to 10 scenarios were added, and the validation was carried out on 100 more unseen scenarios. The models were implemented using the Python Xpress module, and solved using FICO Xpress with relative mip gap set to 10^{-3} , and no timeout.

4 Results & Discussion

The objective of this study was to evaluate how different rebalancing strategies affect bike availability and the resulting carbon emissions caused by users switching to private cars when no bikes are available. The discussion interprets the outcomes of the rebalancing scenarios under three different fleet sizes (160, 180, and 200) and assesses their implications on operational efficiency and emission reduction. We will discuss the results for the following scenarios:

- No rebalancing (available vehicle capacity = 0)
- Limited rebalancing (available vehicle capacity = 12)
- Full rebalancing (available vehicle capacity = Fleet available)

4.1 Model Convergence and Statistical Stability

The simulation methodology involves running the Model optimisation and then validating the resulting optimal rebalancing policy against a growing number of uncertain demand scenarios. Analysing the trend across these scenarios provides insight into the statistical stability and prediction accuracy of the model under uncertainty.

4.1.1 Convergence of Emissions Metrics

Tables 2b, 3b, and 4b show us that as the simulation runs more scenarios (from a single scenario up to 10 scenarios), the model emissions tend towards those obtained by the validation process. This suggests that the model approximates the stochastic space more accurately, even with only 10 sampled scenarios.

- **Initial Variability (Low Scenario Count):** At a low scenario count (e.g., $N = 1$ or $N = 2$), the Validation Emissions often show high deviation and are typically higher than the Model Emissions. This occurs because one or two particularly challenging demand scenarios can skew the average validation emissions upwards.
- **Stabilization (High Scenario Count):** As the number of scenarios increases (e.g., $N > 5$), the Validation Emissions average stabilizes as indicated in figure 1 (a figure for the case with a fleet size of 180). The figure shows that the graph tends to smooth, and the gap between the model emission and validation emission tends to decrease as the number of scenarios is increased, and when we have more rebalancing capacity.
- **Asymptotic Behaviour :** In the most efficient configurations (e.g., 180/180) as seen in Figure 1, the Validation Emissions and Model Emissions may appear to move closer together, indicating that the rebalancing policy is highly robust and performs consistently even when facing a fully explored distribution of demand uncertainty.

This convergence confirms that the overall trends and conclusions drawn from the final averages (case with 10 scenarios) are statistically reliable representations of the system’s expected performance under long-term operational variability.

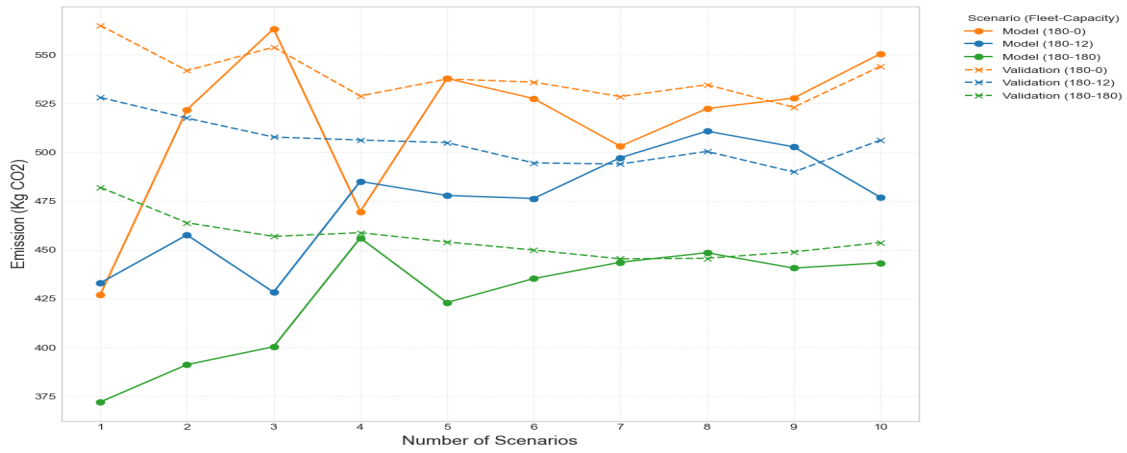


Figure 1: Model and Validation emissions across different numbers of scenarios

4.1.2 Impact on Model Validation Feasibility

Validation Feasibility is the fraction (or percentage) of times the system model successfully found a solution satisfying the minimum commercial demand constraints, when faced with unpredictable, real-world data it hasn’t seen before.

$$\text{Validation Feasibility} = \frac{\text{Number of Successful (Feasible) Validation Runs}}{\text{Total Number of Validation Runs (in this case 100)}}$$

The results shown in Tables 2b, 3b, and 4b show that the Validation Feasibility stabilises as the scenario count increases. Overall, the model is feasible (reaching the goal set as 0.75) under limited or full rebalancing and fails to reach feasibility when no rebalancing is performed and a small amount of fleet is available to meet the fluctuating demand. Figure 2 below shows how the model’s reliability improves as you give it more scenarios to learn from (Model convergence). Another important information from Figure 2 is that Larger fleets (in this case, 200) consistently achieve higher baseline feasibility than smaller fleets (160) across all rebalancing capacities. Additionally, the following trend analysis can be highlighted:

- **Full Rebalance (e.g., 200 Fleet Available and 200 Capacity):** These models are highly robust and achieve 100% feasibility immediately, regardless of the number of training scenarios. The ample rebalancing capacity effectively mitigates demand uncertainty.
- **No Rebalance (Capacity 0):** Smaller fleets (160/0) exhibit extremely low initial feasibility and recover only after being trained on a large scenario. This confirms that without rebalancing, the system has no way to adjust to unexpected demand shifts, meaning the system frequently fails.
- **Limited Capacity (Capacity 12):** These configurations show a strong, converging trend. They begin with a moderate feasibility of 60% to 90% and require additional scenarios to reach stable, high feasibility, indicating that the model learns to allocate limited rebalancing effort more effectively with more data, making it less likely to fail when facing new data.

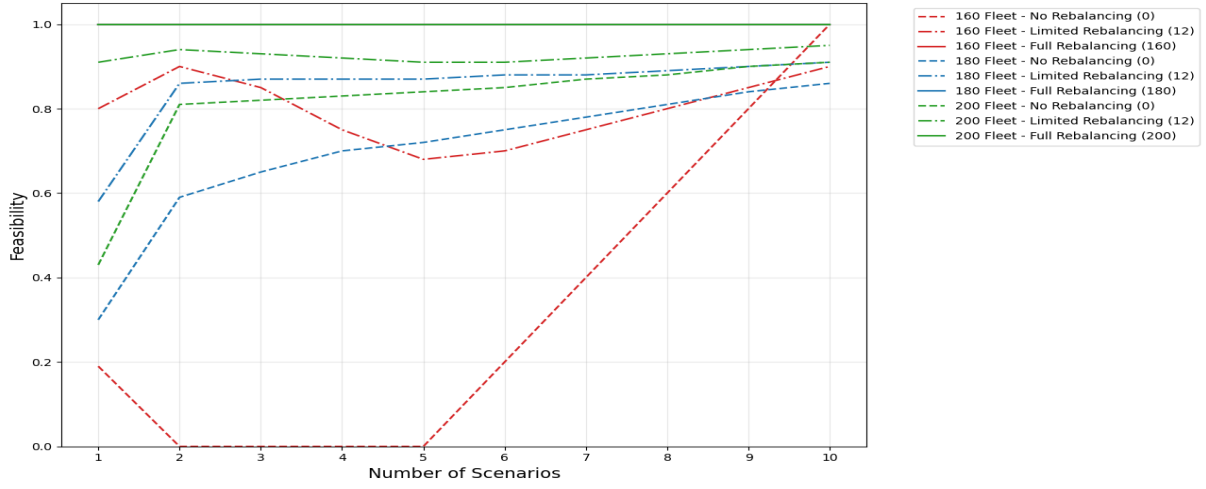


Figure 2: Model Validation Feasibility Convergence

4.2 Analysis of Emissions by Fleet Size

Because model behavior stabilizes at higher scenario counts, the emission and performance metrics analyzed here are taken from scenario number 10, results in Tables 2, 3 and 4, where all fleet–capacity configurations exhibit their most consistent and converged values. This ensures comparability across scenarios and avoids transient early-scenario fluctuations.

4.2.1 Fleet Size 160

No of Scenarios	Difference in car emissions			Average emissions			Average Validation emissions			SD Validation emissions		
	Cap (0)	Cap (12)	Cap (160)	Cap (0)	Cap (12)	Cap (160)	Cap (0)	Cap (12)	Cap (160)	Cap (0)	Cap (12)	Cap (160)
1	-20.43	76.17	62.58	671.13	534.37	457.10	650.69	610.54	519.68	0.00	0.00	0.00
2		26.82	27.65		566.32	468.49		593.14	496.14		8.08	49.85
3		47.27	-2.61		528.38	499.41		575.66	496.80		22.40	30.54
4		30.82	8.67		551.61	475.50		582.43	484.17		40.84	64.52
5		16.97	12.08		566.75	476.65		583.72	488.73		48.31	39.41
6		-18.39	12.77		592.51	475.66		574.12	488.43		38.01	17.21
7	12.44	0.00	21.52	658.23		461.73	670.68		483.25	45.93		36.02
8		5.01	33.82		572.11	458.48		577.12	492.29		64.52	31.53
9	-22.96	32.48	7.04	660.03	539.76	483.31	637.07	572.24	490.35	55.36	61.99	44.61
10	5.94	15.62	32.42	682.71	557.13	456.38	688.65	572.75	498.80	55.44	55.44	45.30

(a) Car Emissions Data (Emissions in kg CO₂) where SD is the standard deviation, and Cap is the rebalancing capacity. The difference in car emissions is between Average Validation emissions and Average emissions. The empty rows shows that the base model was infeasible when certain number of scenarios are considered.

No of Scenario	Average Van emissions			Average Van Validation emissions			Validation feasibility		
	Capacity (0)	Capacity (12)	Capacity (160)	Capacity (0)	Capacity (12)	Capacity (160)	Capacity (0)	Capacity (12)	Capacity (160)
1	0.00	6.95	30.85	0.00	8.25	34.35	0.19	0.80	1.00
2		6.44	39.36		8.22	36.62		0.83	1.00
3		7.16	37.63		7.96	33.75		0.67	1.00
4		7.16	31.84		7.50	35.98		0.74	1.00
5		7.14	33.77		8.80	36.00		0.68	1.00
6		7.90	35.21		8.02	37.10		0.74	1.00
7	0.00		33.27	0.00		35.72	0.62		1.00
8		6.58	33.30		7.41	37.19		0.79	1.00
9	0.00	7.19	30.89	0.00	7.55	35.74	0.48	0.57	1.00
10	0.00	7.45	32.40	0.00	7.55	35.10	0.64	0.70	1.00

(b) Van Emissions Data and Validation (Emissions in kg CO₂).

The empty rows shows that the base model was infeasible when certain number of scenarios are considered.

Table 2: Iteration results with fleet of 160 Bikes.

- **No Rebalancing (Capacity = 0):** The results from Table 2a shows that with no rebalancing, the system cannot correct supply–demand imbalances, resulting in extremely low feasibility (0.19). This forces many users to switch to cars, which explains the very high model car emissions (682.71 kg CO₂) that almost match the validation emissions. Even though no rebalancing was done, and thus the van emissions were zero, the overall environmental impact is still poor because the system repeatedly fails to meet demand. Essentially, a lack of rebalancing makes service failure the default, leading to high total emissions.

- **Limited Rebalancing (Capacity = 12):** Introducing a small rebalancing capacity significantly improves performance. Van emissions remain low (7.45 kg CO₂) as seen in Table 2b, but they enable a large reduction in car emissions from 682.71 kg CO₂ to 557.13 kg CO₂ (≈ 125 kg of CO₂ saved). This demonstrates that even minimal rebalancing can prevent the worst shortages and yield substantial environmental benefits. However, feasibility (0.68) remains below the acceptable threshold (0.75), meaning the system is improving but still unreliable.
- **Full Rebalancing (Capacity = 160):** Full operational flexibility results in perfect validation feasibility (1.00) as shown in Table 2b, allowing the system to meet demand requirements across all scenarios. Although van emissions increase to 32.4 kg CO₂, this is more than offset by the substantial reduction in car emissions (456.38 kg CO₂). This scenario provides the best environmental and operational performance, showing that investing in rebalancing yields a reliable and low-emission system.

4.2.2 Fleet Size 180

No of Scenarios	Difference in car emissions			Average emissions			Average Validation emissions			SD Validation emissions		
	Cap (0)	Cap (12)	Cap (180)	Cap (0)	Cap (12)	Cap (180)	Cap (0)	Cap (12)	Cap (180)	Cap (0)	Cap (12)	Cap (180)
1	104.88	93.05	137.95	427.04	427.59	341.50	564.99	520.64	446.38	51.88	44.68	46.21
2	72.78	57.04	20.08	521.79	452.43	354.73	541.87	509.47	427.51	44.59	56.15	36.93
3	52.62	78.46	-9.40	563.25	422.16	369.69	553.85	500.62	422.31	48.42	51.15	39.07
4	4.11	20.38	59.24	469.51	478.22	419.57	528.85	498.60	423.78	46.27	49.44	36.55
5	26.44	26.48	-0.32	537.85	471.63	393.41	537.53	498.11	419.85	49.62	52.96	41.15
6	12.93	18.62	8.45	527.52	468.42	401.42	535.97	487.04	414.35	50.28	51.07	35.70
7	-1.30	-4.00	25.45	503.09	490.22	411.45	528.54	486.22	410.15	50.13	47.27	34.01
8	-4.70	-10.27	12.24	522.38	503.20	414.79	534.62	492.93	410.09	46.05	58.93	35.36
9	7.43	-13.54	-4.73	527.76	495.94	407.84	523.03	482.39	415.27	41.70	51.78	36.27
10	11.93	28.72	-6.42	550.43	470.29	408.09	544.01	499.01	420.02	46.03	51.52	41.14

(a) Car Emissions Data (Emissions in kg CO₂) where SD is the standard deviation, and Cap is the rebalancing capacity. The difference in car emissions is between Average Validation emissions and Average emissions.

No of Scenario	Average Van emissions			Average Van Validation emissions			Validation feasibility		
	Capacity (0)	Capacity (12)	Capacity (180)	Capacity (0)	Capacity (12)	Capacity (180)	Capacity (0)	Capacity (12)	Capacity (180)
1	0.00	5.44	30.56	0.00	7.47	35.66	0.30	0.58	1.00
2	0.00	5.23	36.46	0.00	8.04	36.39	0.59	0.70	1.00
3	0.00	6.16	30.68	0.00	7.17	34.60	0.56	0.74	1.00
4	0.00	6.85	36.43	0.00	7.65	35.03	0.36	0.78	1.00
5	0.00	6.24	29.59	0.00	6.86	34.20	0.72	0.85	1.00
6	0.00	7.91	33.87	0.00	7.52	35.57	0.59	0.88	1.00
7	0.00	6.98	32.20	0.00	7.75	35.25	0.58	0.96	1.00
8	0.00	7.62	33.74	0.00	7.51	35.61	0.57	0.86	1.00
9	0.00	6.88	32.82	0.00	7.52	33.67	0.53	0.86	1.00
10	0.00	6.66	35.23	0.00	7.16	33.76	0.67	0.84	1.00

(b) Van Emissions Data and Validation (Emissions in kg CO₂.)

Table 3: Iteration results with fleet of 180 Bikes.

- **No Rebalancing (Capacity = 0):** With a larger fleet, the system passively handles demand fluctuations much better, reaching a feasibility of 0.67 as can be seen in Table 3b. Also, Table 3a shows that car emissions drop to 550.43 kg CO₂, showing that increasing fleet size alone can substantially cut reliance on cars even without rebalancing.
- **Limited Rebalancing (Capacity = 12):** Again, adding a small rebalancing capacity raises feasibility to 0.84, moving the system close to fully reliable. This comes at a very low operational cost (6.66 kg CO₂ in van emissions) as seen in Table 3b while producing a notable reduction in car emissions to 470.29 kg CO₂. This scenario provides an efficient balance between operational effort and environmental benefits.
- **Full Rebalancing (Capacity = 180):** Full rebalancing yields the strongest performance, minimizing car emissions to 408.09 kg CO₂ (see Table 3a). Although van emissions increase to 35.23 kg CO₂, the reduction in car emissions far outweighs this cost. This configuration delivers the best environmental outcome and ensures robust service across all demand conditions.

4.2.3 Fleet Size 200

- **No Rebalancing (Capacity = 0):** At this scale, the fleet is inherently reliable (Feasibility 0.85), but still environmentally inefficient. Car emissions remain relatively high at 459.29 kg CO₂ as seen in Table 4a, indicating that passive reliability alone does not minimize user car dependence without active rebalancing.
- **Limited Rebalancing (Capacity = 12):** With only a small rebalancing effort, the system reaches near-perfect feasibility (0.91) while keeping van emissions extremely low (6.86 kg CO₂) as seen in Table 4b. Car emissions drop to 407.13 kg CO₂, making this scenario an effective low-cost, high-impact operational strategy.

No of Scenarios	Difference in car emissions			Average emissions			Average Validation emissions			SD Validation emissions		
	Cap (0)	Cap (12)	Cap (200)	Cap (0)	Cap (12)	Cap (200)	Cap (0)	Cap (12)	Cap (200)	Cap (0)	Cap (12)	Cap (200)
1	46.21	51.68	104.98	370.29	382.09	287.00	501.41	453.27	397.88	0.00	0.00	0.00
2	46.54	51.70	96.50	470.48	386.92	337.22	496.34	454.20	379.29	41.07	4.95	27.03
3	46.31	48.92	90.05	480.86	428.24	353.44	492.17	433.91	374.19	80.89	22.13	23.10
4	45.50	52.01	92.43	480.06	399.14	338.27	481.72	434.56	380.87	22.87	29.55	20.65
5	45.57	51.15	94.26	448.62	437.01	381.95	490.15	431.89	364.77	28.62	30.11	11.57
6	45.21	51.23	92.25	445.67	396.75	352.31	471.84	433.36	367.45	56.83	28.60	15.97
7	45.21	51.23	92.25	451.05	413.45	352.63	477.68	433.80	369.63	16.63	37.96	38.52
8	46.30	47.92	90.17	442.86	433.68	361.98	473.83	431.04	366.87	42.39	34.41	16.57
9	43.10	49.21	97.56	442.74	414.95	364.94	465.63	418.33	366.03	42.04	27.09	17.10
10	41.14	51.35	46.39	459.29	407.13	359.31	480.72	440.24	364.73	42.43	38.38	33.82

(a) Car Emissions Data (Emissions in kg CO₂) where SD is the standard deviation, and Cap is the rebalancing capacity. The difference in car emissions is between Average Validation emissions and Average emissions.

No of Scenario	Average Van emissions			Average Van Validation emissions			Validation feasibility		
	Capacity (0)	Capacity (12)	Capacity (200)	Capacity (0)	Capacity (12)	Capacity (200)	Capacity (0)	Capacity (12)	Capacity (200)
1	0	5.44	30.56	0	7.47	36.96	0.43	0.91	1
2	0	4.62	30.68	0	9.06	36.54	0.81	0.94	1
3	0	6.00	30.60	0	5.73	36.39	0.89	0.99	1
4	0	4.91	29.99	0	7.25	35.62	0.73	0.90	1
5	0	5.41	30.33	0	7.15	35.86	0.84	0.91	1
6	0	4.94	30.52	0	5.75	35.93	0.60	0.91	1
7	0	5.68	30.44	0	7.75	35.25	0.76	0.94	1
8	0	5.77	30.47	0	7.52	35.45	0.73	0.97	1
9	0	6.00	30.53	0	7.52	35.33	0.78	0.99	1
10	0	6.86	32.53	0	7.16	36.70	0.85	0.79	1

(b) Van Emissions Data and Validation (Emissions in kg CO₂).

Table 4: Iteration results with fleet of 200 Bikes.

- **Full Rebalancing (Capacity = 200):** Full rebalancing delivers the best environmental performance. Car emissions reach the minimum value of 359.31 kg CO₂ as seen in Table 4a, while van emissions (34.03 kg CO₂) are lower than those of the smaller fleets. This indicates that larger fleet density improves logistical efficiency and rebalancing becomes easier, cheaper, and cleaner as fleet size increases. This supports investment in both higher fleet capacity and flexible rebalancing policies.

The Figure 3 below shows how total emissions change as the model is trained on an increasing number of scenarios for different fleet sizes (160, 180, 200) and rebalancing capacities (0, 12, Full). The emission results show that as the number of training scenarios increases, the model's behaviour converges and emissions stabilise. Small fleets—particularly the 160 bike system without rebalancing exhibit high and unstable emissions, reflecting their vulnerability to inadequate supply and the inability to reposition bikes. Increasing rebalancing capacity consistently reduces emissions, with full rebalancing producing the cleanest and most stable outcomes. Larger fleets (200 bikes) show both lower emissions and smoother convergence across scenarios, highlighting their greater resilience to demand uncertainty. By Scenario 10, all configurations have stabilised, making it the most reliable point of comparison. The dotted line segments in figure 3 represent scenario points where the optimisation model failed to converge (it was infeasible), and therefore no valid emission output was generated.

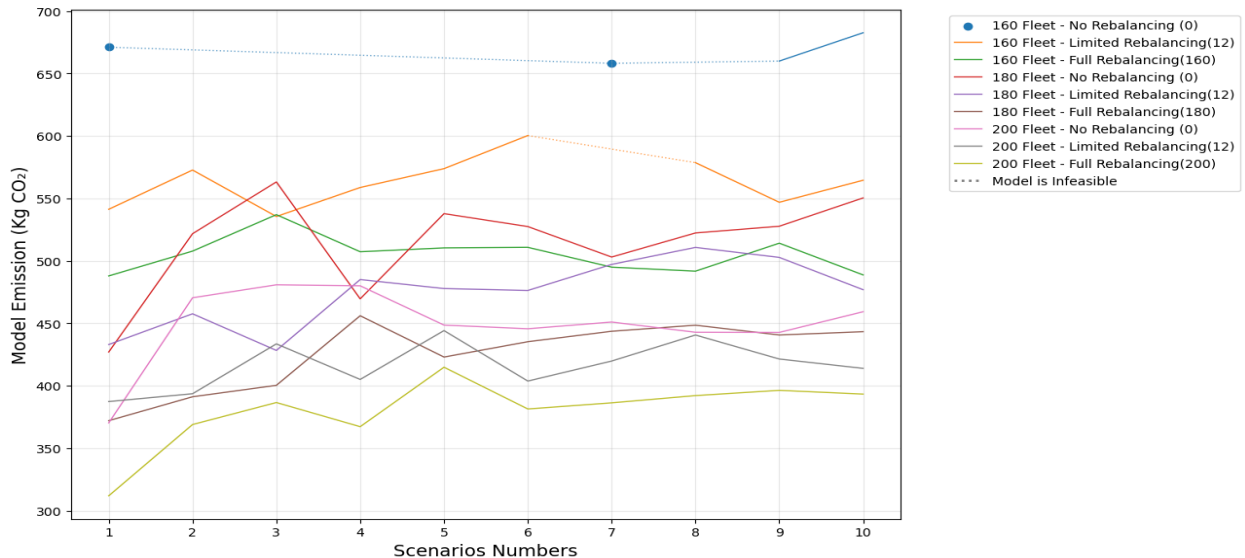
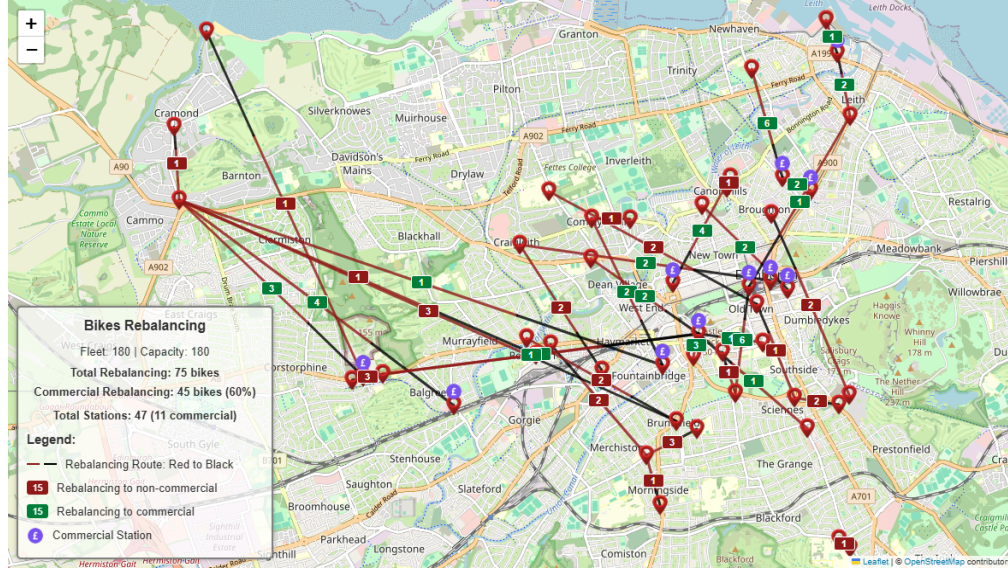


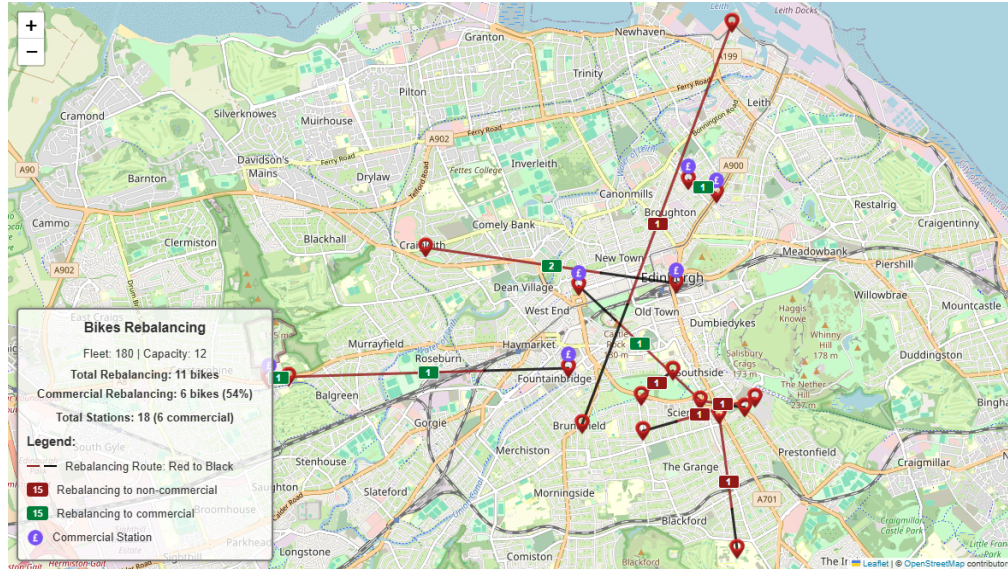
Figure 3: Total average model emissions across different scenario numbers

4.3 Effect of Rebalancing on Bike Allocation

Across all fleet sizes, the allocation maps show that higher rebalancing capacity progressively smooths spatial imbalances by shifting bikes from consistently oversupplied outer stations toward high-demand central zones, resulting in a more uniform and reliable distribution across the network. Figures 4 below show a sample of how the model does the allocation across different locations when the fleet available is 180 bikes, when the limited and the full rebalancing is performed (capacity = 12 and 180). In the limited vehicle capacity scenario, only 11 bikes were rebalanced, with 54% directed to commercial stations, affecting 18 stations overall. In contrast, under full capacity, 75 bikes were rebalanced, 60% toward commercial stations, covering 47 stations. The output shows that while commercial stations receive priority to meet the service constraint, non-commercial stations are also rebalanced, ensuring broader service coverage. The fact that only 75 of the 180 bikes required repositioning suggests that a large portion of the fleet was already well distributed to meet demand.



(a) Rebalancing with fleet=180, and rebalancing capacity=180



(b) fleet=180, and rebalancing capacity=12

Figure 4: Bike Rebalancing Allocation Maps for fleet of 180 bikes and rebalancing capacity of 180 and 12, respectively

4.4 Value of perfect information

This section examines the performance differences between the model emissions, with perfect foresight (Perfect Information) and Non-Perfect Information. The first scenario created during the sampling was chosen as the correct one. Table 5 below shows the values that represent the loss (in terms of kg CO₂) incurred when the system relies on NPI instead of PI. Only the values that resulted in a validation success (feasibility > 75%) were presented.

As we can see, increasing the van capacity lowers the disparity between the situation with perfect foresight and a non-perfect one. This suggests that a system with more rebalancing capacity is more robust to making suboptimal decisions,

Number of scenarios	Fleet Size(160)		Fleet Size(180)		Fleet Size(200)		
	Van capacity(12)	Van capacity(160)	Van capacity(12)	Van capacity(180)	Van capacity(0)	Van capacity(12)	Van capacity(200)
10	44.4343	25.9467	35.1056	25.9962	39.6957	31.4918	29.9527

Table 5: The difference in total emissions between Perfect and Non-Perfect Information settings

as the daily rebalancing can be performed to compensate accordingly. This can be extrapolated to other scenarios that the model has not seen. On the other hand, we cannot conclude that increasing the fleet capacity makes the model more resilient to suboptimal decisions.

5 Conclusions

We have proposed an optimisation model to find an optimal daily rebalancing and bike allocation plan that would decrease the CO2 emissions in Edinburgh, while keeping the rates of demands satisfied in certain privileged (e.g, commercial) areas. The model had to return highly sustainable plans, despite uncertain demands, while not compromising the solution’s robustness and computational efficiency. To achieve it, we have employed an out-of-sample validation scheme.

We evaluate how different rebalancing strategies and fleet sizes affect the model’s robustness, bike availability, carbon emissions, and operational efficiency, focusing on three scenarios: no, limited, and full rebalancing for fleet sizes of 160, 180, and 200 bikes.

As expected, when the model is supplied with a greater number of scenarios, its feasibility and goal function slowly approach those of the validation scheme. However, it turns out that the rebalancing policy has a great impact on the rate of convergence of those metrics, suggesting that the validation process with rebalancing is more forgiving of poor initial allocations. Moreover, increasing the rebalancing capacity can lower the value of perfect information, showing that the expected emissions can approximate the real-life outcomes to a greater extent, as a system with greater rebalancing flexibility is more robust to suboptimal bike initial distribution. On the other hand, while fleet size reduces the total emissions, its effect on robustness is less meaningful.

When reallocating bikes, stations in commercial areas are given priority and allocated the most bikes, due not only to the higher demand but also to the need to satisfy the minimum threshold of total demand. Stations in the non-commercial areas are also rebalanced to ensure more demand is met. The rebalancing allocation maps for the 180-bike fleet confirm that rebalancing prioritises commercial stations to meet service constraints while also supporting non-commercial stations. However, the full rebalancing capacity is rarely needed, suggesting that the system can operate efficiently with only a few rebalancing vans.

When there is no rebalancing, people do switch to cars for commuting, which in turn increases emissions. Increasing rebalancing, even a little, can drastically reduce emissions, as not only do the car emissions dominate the resulting van emissions, but also, more demand can be met.

While our findings can give the council an incentive to invest more in the bike infrastructure, we have to acknowledge that our experiments had inherent limitations. First, we have to reconcile with the fact that the proper routing scheme should be implemented – which would not be possible in our case, as we were lacking computational power. Moreover, our model could use a more intricate demand-scenario simulation. Further research could also incorporate a model when some people prefer to ride either a bus or share a car with others when they cannot ride a bike.

Additional overarching limitations include issues of inclusion. We have to consider the physical conditions of all people, including those with disabilities and the elderly, who cannot take a bike, but rather have to use a car. We also didn’t consider non-emissive, electric cars and vans. Additionally, bike rental systems rarely offer an option for carrying cargo, thus excluding people who engage in heavy, retail shopping. Those people could also contribute to the commercial area demands.

6 Recommendations

To construct a green, sustainable city, the Edinburgh City Council can use the model, adding more scenarios and constraints, as it can accommodate the rebalancing problem, station allocation and fleet size needed for the reduction of emissions.

The Edinburgh City Council should approve an investment for an initial fleet size of 180 bikes, coupled with the immediate implementation of at least the limited daily rebalancing policy. This strategy will drastically reduce emissions and maintain high demand satisfaction in a robust and computationally efficient manner. After researching the impact of this scheme in practice, the council might take into consideration either expanding the fleet size or buying more rebalancing vans, the latter being given a priority.